

Enterprise Modernization Intelligence Report

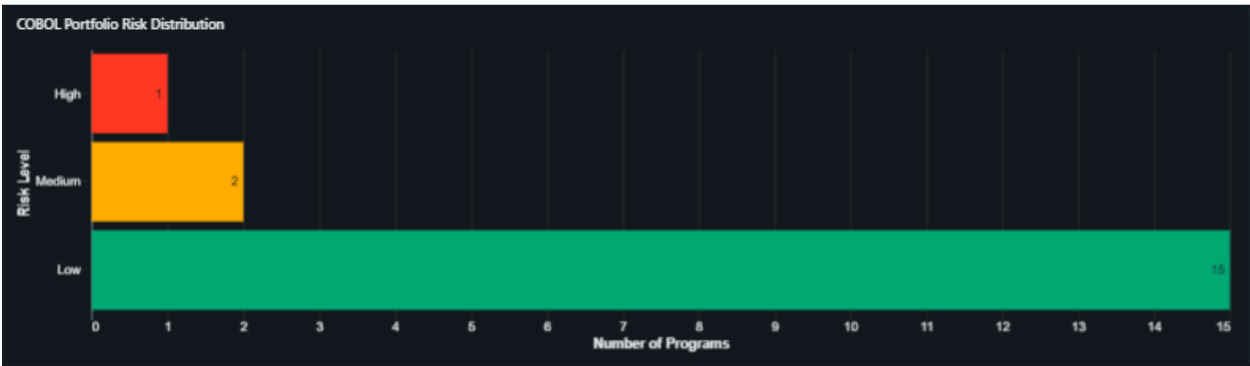
Executive Summary

Your mainframe portfolio contains **18 COBOL programs** with a **favorable modernization outlook**: 83% (15 programs) are low-risk API candidates, while 3 programs require phased modernization planning due to complex DB2 dependencies and external program calls.

Critical Finding: PAYROLL.cbl is your highest-risk asset (readiness score: 40) with 5 governance alerts, requiring immediate attention before migration.

1. COBOL Program Inventory

Your portfolio includes 18 COBOL programs ranging from 3 to 78 lines of code.



Risk Breakdown:

- **1 High-Risk Program** (PAYROLL.cbl): Complex dependencies, copybook usage, DB2 write operations
- **2 Medium-Risk Programs** (CUSTOMER.cbl, FUNDSXFER.cbl): DB2 dependencies with UPDATE operations
- **15 Low-Risk Programs:** Minimal dependencies, excellent API candidates



2. Critical Modernization Risks & Governance Alerts

5 Active Governance Alerts across 4 assets require immediate attention:

CRITICAL Severity (2 alerts)

- **JOB001.jcl**: Requires phased modernization planning
- **PAYROLL.cbl**: Requires phased modernization planning

HIGH Severity (3 alerts - DB2 Operations)

- **CUSTOMER.cbl**: Performs critical DB2 UPDATE operations on CUSTOMER_MASTER
- **FUNDSXFER.cbl**: Performs critical DB2 UPDATE operations on ACCOUNT_MASTER
- **PAYROLL.cbl**: Performs critical DB2 UPDATE operations on EMP_MASTER and PAYROLL_AUDIT

3. DB2-Dependent Files

4 files have DB2 dependencies with a total of 7 database operations:

File	Type	Operations	Tables Accessed	Operation Count
PAYROLL.cbl	CBL	SELECT, UPDATE, INSERT	EMP_MASTER, PAYROLL_AUDIT	3
CUSTOMER.cbl	CBL	SELECT, UPDATE	CUSTOMER_MASTER	2
FUNDSXFER.cbl	CBL	UPDATE	ACCOUNT_MASTER	1
MONTHEND_RECON.sql	SQL	SELECT	PAYMENT_AUDIT	1

Migration Impact: All three COBOL programs with DB2 write operations require database migration strategy and transaction integrity validation.

4. JOB001 Batch Flow Analysis

JOB001.jcl orchestrates a 4-step batch process with 7 datasets:

Execution Flow:

1. **SORTSTEP** (UTILITY): SORT utility for data preparation
2. **PAYSTEP** (APPLICATION): Executes PAYROLL.cbl program
3. **CUSTSTEP** (APPLICATION): Executes CUSTOMER.cbl program
4. **RPTSTEP** (APPLICATION): Executes REPORTGEN program

Key Datasets:

- Input: EMP.RAW.INPUT.FILE, EMP.INPUT.FILE, CUSTOMER.INPUT.FILE
- Output: EMP.OUTPUT.FILE, PAYROLL.DAILY.REPORT
- Libraries: PAYROLL.LOADLIB, CUSTOMER.LOADLIB

Modernization Recommendation: Convert this JCL flow into cloud-native workflow orchestration (e.g., Apache Airflow, Azure Data Factory, AWS Step Functions) with containerized program execution.

5. Modernization Readiness Summary

Modernization Readiness Summary						
Risk Level	Priority	API Candidate	Program Count	Avg Readiness	Total LOC	
High	High	NO	1	40.0	78	
Medium	Medium	NO	2	50.0	67	
Low	Low	YES	15	99.0	71	

Portfolio Health Metrics:

- **Average Readiness Score:** 87.2 (weighted across all programs)
- **API Candidates:** 15 programs (83%) ready for immediate API enablement
- **Total Code Volume:** 216 lines across all COBOL programs

Quick Wins: The 15 low-risk programs with 100% readiness scores can be migrated first to build momentum and establish patterns.

6. Migration Blockers

3 programs present migration complexity requiring detailed planning:

High Priority: PAYROLL.cbl

- **Risk Factors:** External program dependency (EMPREAD, TAXCALC), Copybook dependency (EMPREC), DB2 operations, Database write operations
- **Dependencies:** Calls 2 external programs, uses 1 copybook
- **Recommendation:** Needs detailed impact analysis, phased modernization, and regression validation

Medium Priority: CUSTOMER.cbl & FUNDSXFER.cbl

- **CUSTOMER.cbl:** Calls CUSTAUDIT, performs DB2 SELECT/UPDATE on CUSTOMER_MASTER
- **FUNDSEXFER.cbl:** Calls FRAUDCHK, performs DB2 UPDATE on ACCOUNT_MASTER
- **Recommendation:** Requires dependency, DB2, and batch flow review before modernization

Dependency Analysis:

- PAYROLL.cbl has the most complex dependency graph (2 CALL dependencies + 1 COPYBOOK)
- All three blockers have external program dependencies that must be migrated or stubbed

7. Business Rules Extracted from COBOL

25 business rules identified across COBOL programs with varying confidence levels:

High-Confidence Rules (6 rules from PAYROLL.cbl & CUSTOMER.cbl)

PAYROLL.cbl Business Logic:

- **Payroll Tax Rule:** Employees with salary > \$50,000 are assigned higher tax amounts
- Recommendation: Externalize tax threshold into configurable rule table
- **Net Salary Calculation:** Net salary = Salary - Tax amount
- Recommendation: Expose as API or rules engine service
- **Audit Trail:** Creates payroll audit records for tracking
- Recommendation: Preserve audit trail in target cloud platform
- **Employee Master Data:** Reads/updates EMP_MASTER for payroll details
- Recommendation: Map to employee domain data model

CUSTOMER.cbl Business Logic:

- **Customer Master Data:** Reads/updates CUSTOMER_MASTER details
- Recommendation: Map to customer domain data model
- **Customer Audit:** Calls CUSTAUDIT for customer risk tracking
- Recommendation: Review downstream audit dependency before API decomposition

Medium-Confidence Rules (4 rules)

- DB2 data processing patterns identified across CUSTOMER.cbl, FUNDSEXFER.cbl, and PAYROLL.cbl
- Recommendation: Include in DB2 lineage and data migration assessment

Low-Confidence Rules (15 rules)

- Legacy processing patterns across 15 programs (ACCOUNT.cbl, ACCTOPEN.cbl, AUTHCTRL.cbl, etc.)
- Recommendation: Requires manual review for modernization scope

Strategic Recommendations

Phase 1: Quick Wins (Weeks 1-4)

1. **Migrate 15 low-risk API candidates** with 100% readiness scores
2. **Establish cloud-native patterns** for stateless COBOL programs
3. **Build CI/CD pipeline** for automated testing and deployment

Phase 2: Medium Complexity (Weeks 5-12)

1. **Modernize CUSTOMER.cbl and FUNDSXFER.cbl** with DB2 migration strategy
2. **Implement database abstraction layer** for DB2 dependencies
3. **Convert JOB001.jcl** to cloud-native workflow orchestration

Phase 3: High Complexity (Weeks 13-20)

1. **Phased migration of PAYROLL.cbl** with comprehensive impact analysis
2. **Externalize business rules** (tax thresholds, calculation logic) to rules engine
3. **Validate audit trail preservation** and regulatory compliance
4. **Regression testing** across all integrated systems

Governance & Risk Mitigation

- **Address 5 critical alerts** before migration begins
- **Establish DB2 data migration runbook** for 3 programs with write operations
- **Document and test external program dependencies** (CUSTAUDIT, FRAUDCHK, EMPREAD, TAX-CALC)
- **Preserve copybook schema** (EMPREC) in target data contracts

Next Steps

1. **Prioritize PAYROLL.cbl analysis:** Conduct detailed dependency mapping and impact assessment
2. **Validate DB2 migration approach:** Test transaction integrity for UPDATE/INSERT operations
3. **Prototype JCL-to-workflow conversion:** Prove out orchestration pattern with JOB001.jcl
4. **Engage business stakeholders:** Validate extracted business rules and tax calculation logic
5. **Establish modernization metrics:** Track readiness scores, migration velocity, and defect rates